

# Assignment 4

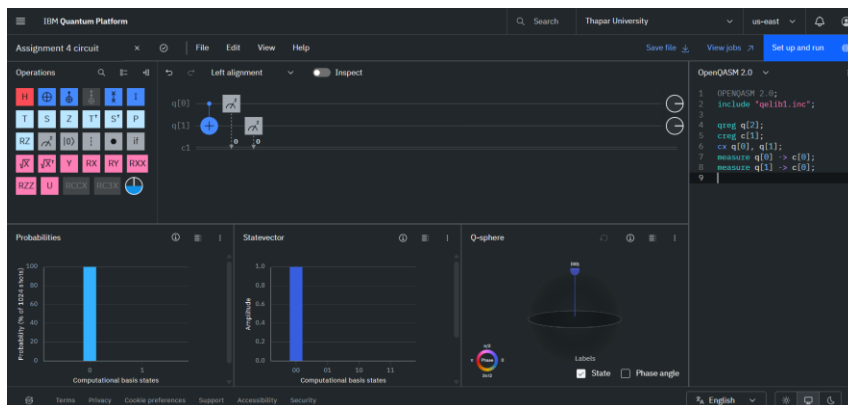
Divjot Singh Gulati 102316025

## Experiment 1-:

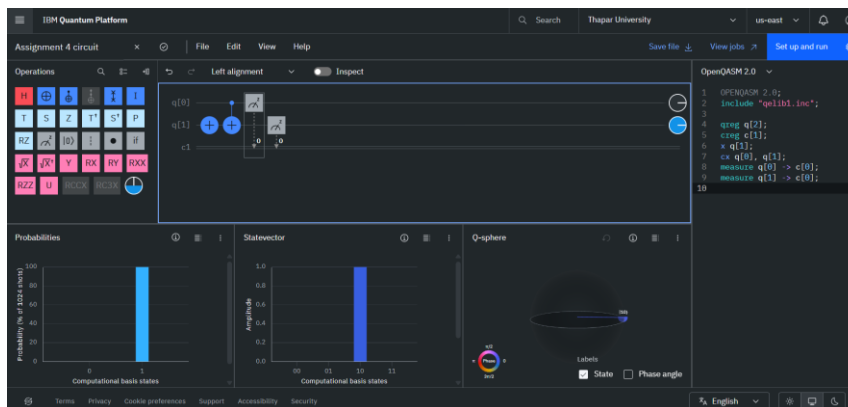
To study the behavior of a two-qubit quantum system and to understand the operation of all fundamental two-qubit gates, including controlled and uncontrolled gates.

### 1. CNOT Gate

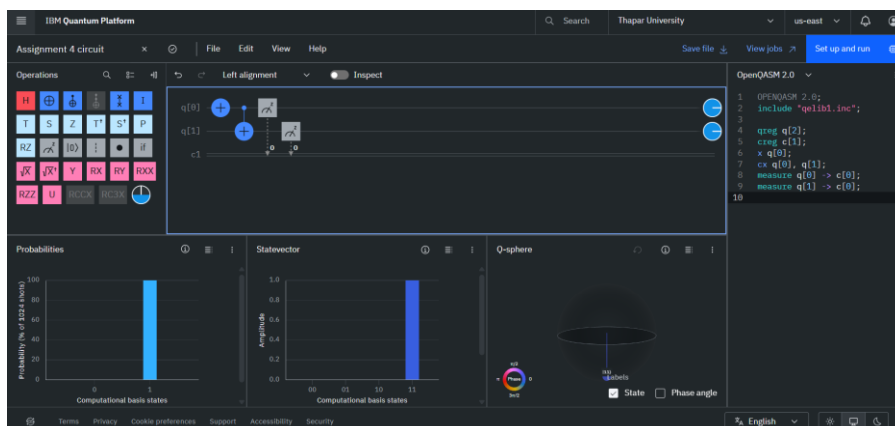
#### a. State $|00\rangle$



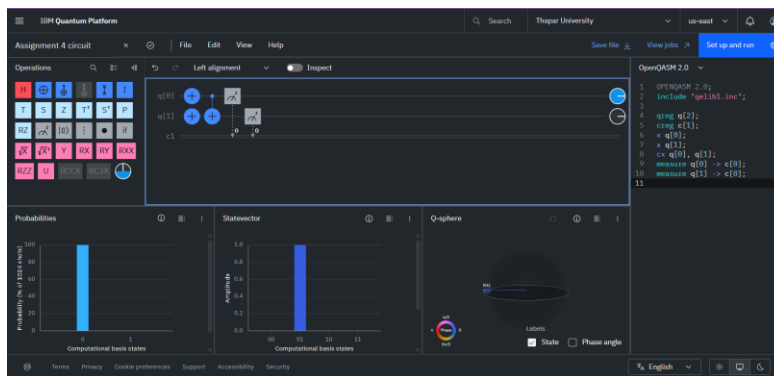
#### b. State $|01\rangle$



#### c. State $|10\rangle$

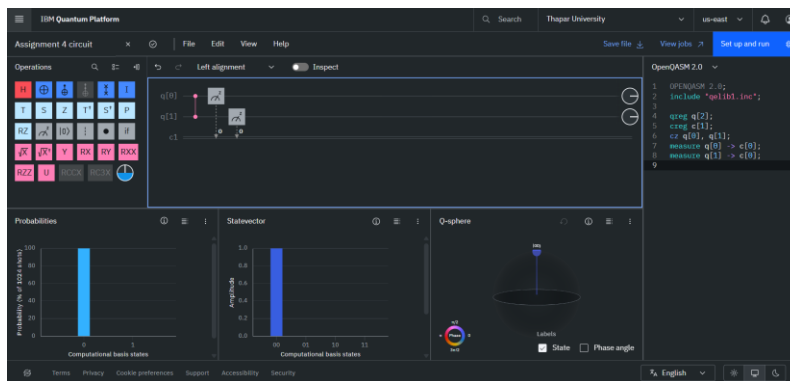


d. State  $|11\rangle$

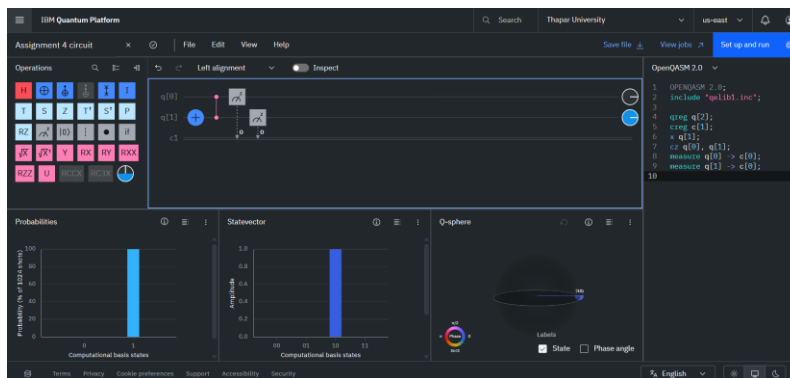


## 2. CZ Gate

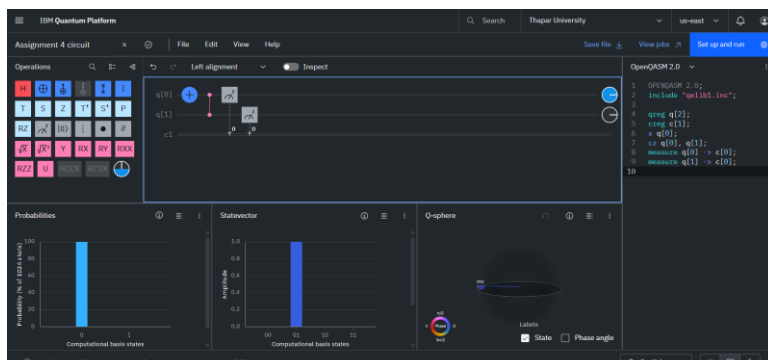
a. State  $|00\rangle$



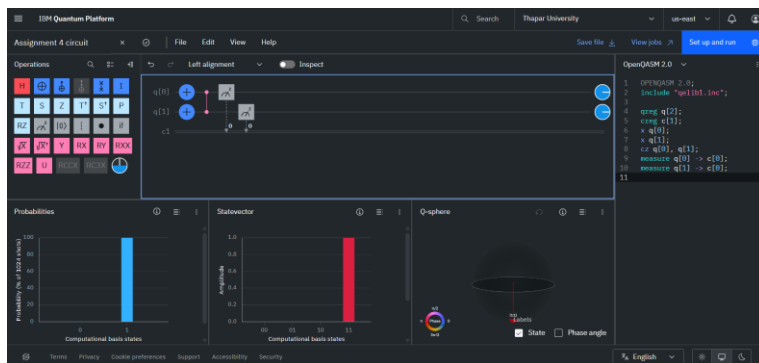
b. State  $|01\rangle$



c. State  $|10\rangle$

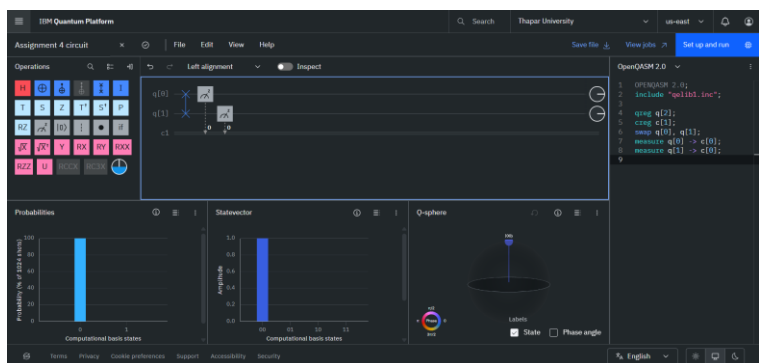


d. State  $|11\rangle$

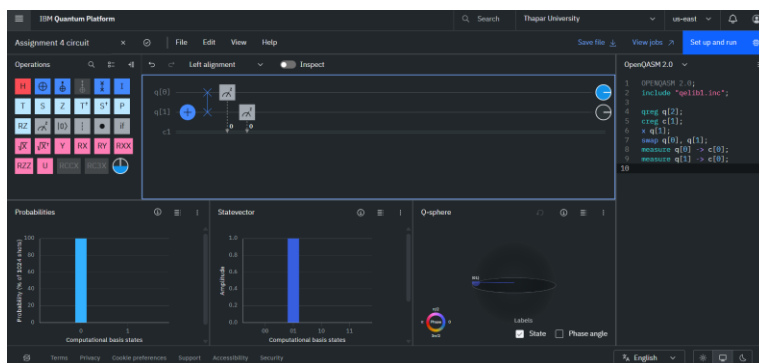


### 3. SWAP Gate

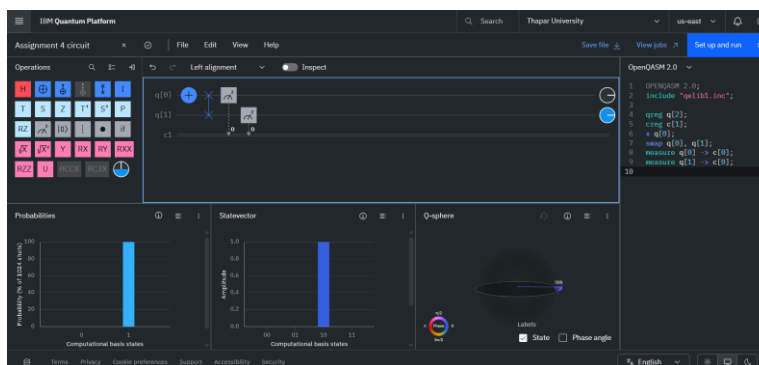
a. State  $|00\rangle$



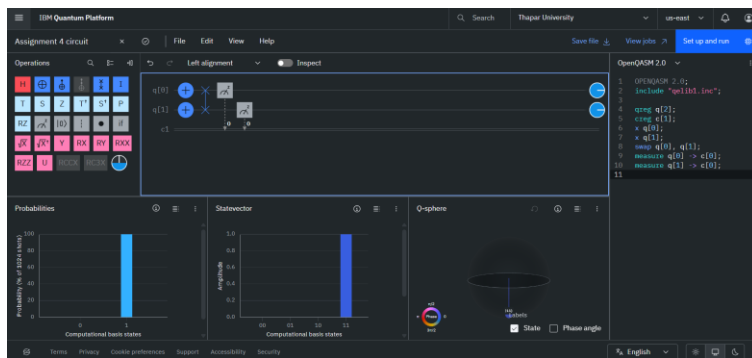
b. State  $|01\rangle$



c. State  $|10\rangle$

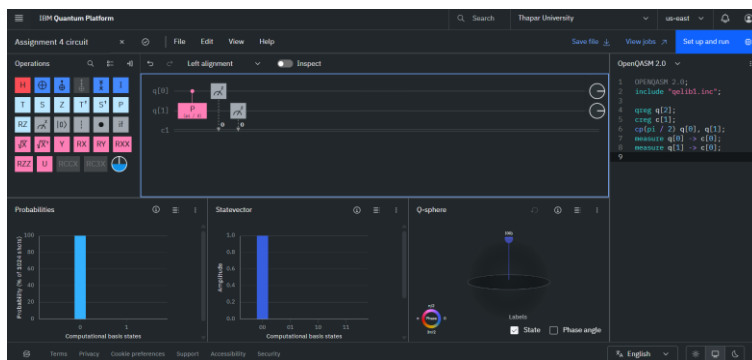


d. State  $|11\rangle$

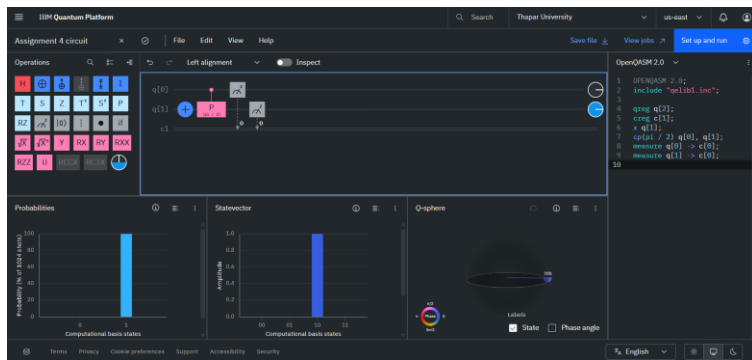


#### 4. Controlled-Phase gate CP( $\phi$ )

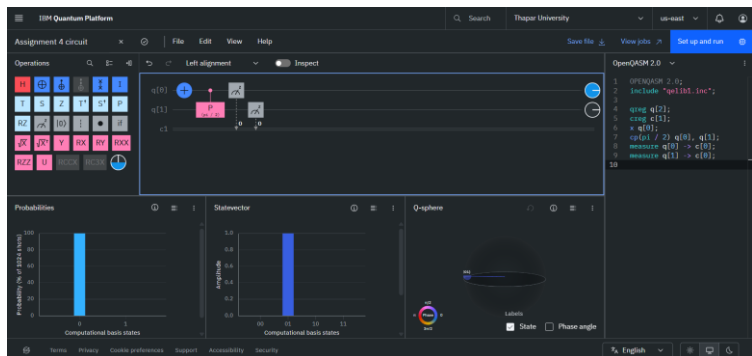
a. State  $|00\rangle$



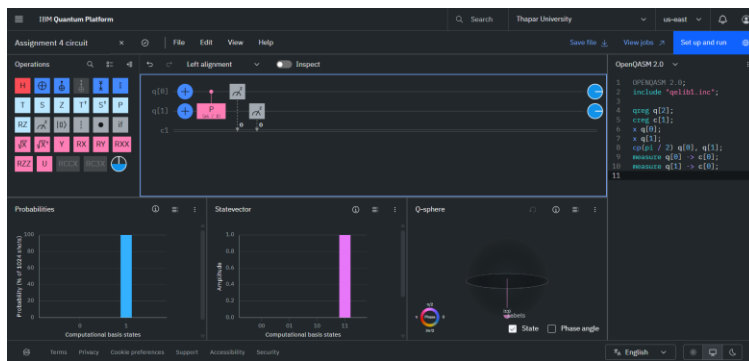
b. State  $|01\rangle$



c. State  $|10\rangle$



d. State  $|11\rangle$

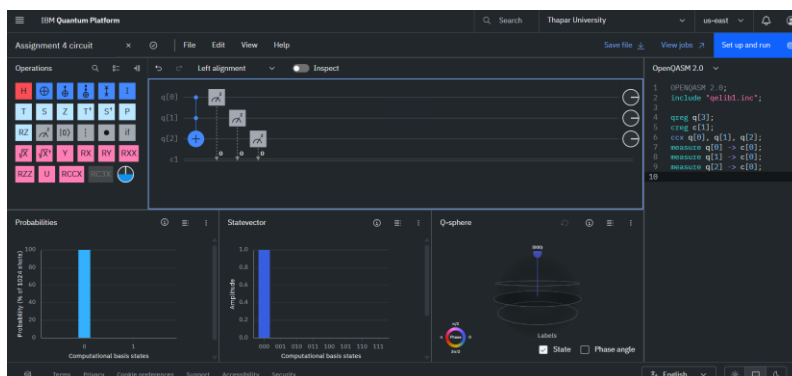


## Experiment 2:-

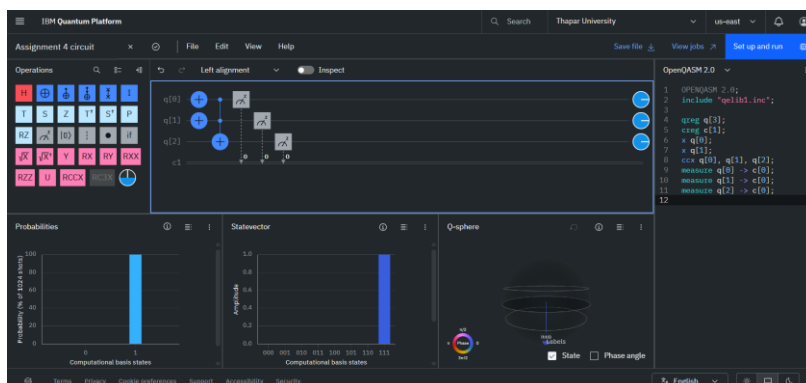
To study the behavior of a three-qubit quantum system and to understand the operation of all fundamental three-qubit gates, including controlled and uncontrolled gates

1. Toffoli gate (CCNOT)

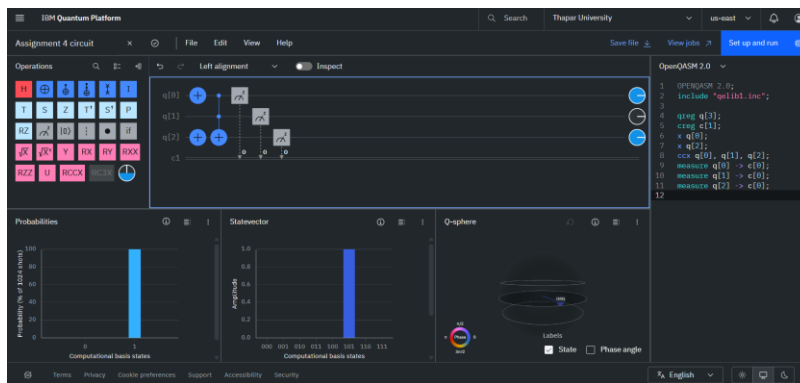
a. State  $|000\rangle$



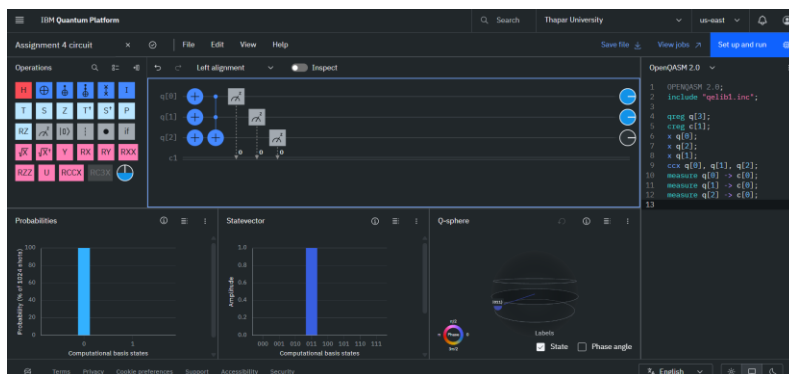
b. State  $|110\rangle$



c. State  $|101\rangle$

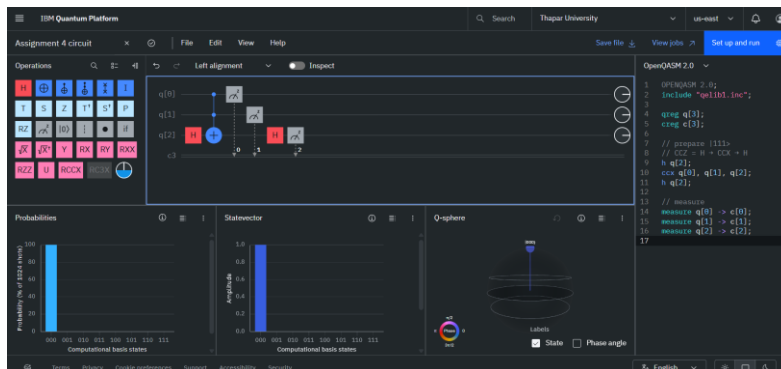


d. State  $|111\rangle$

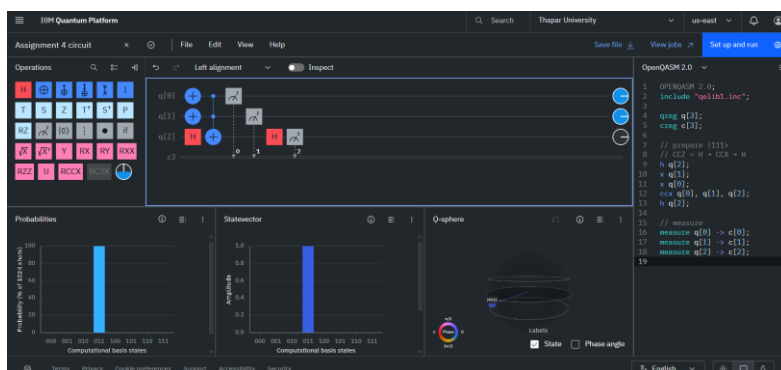


## 2. Controlled-Controlled-Z (CCZ) gate

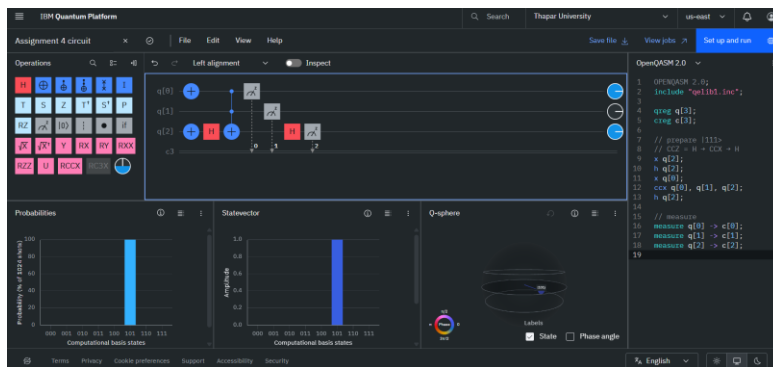
a. State  $|000\rangle$



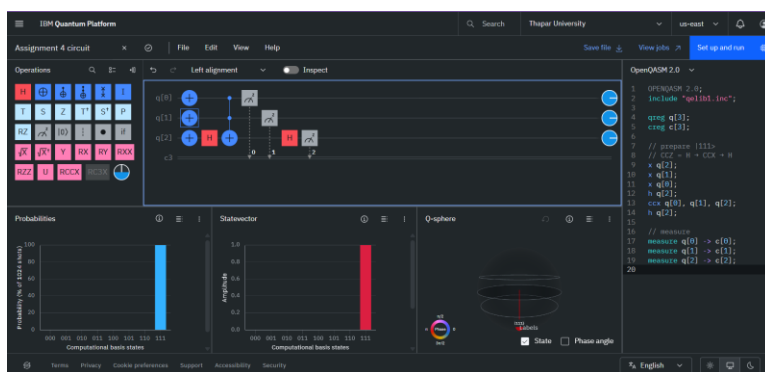
b. State  $|110\rangle$



c. State  $|101\rangle$

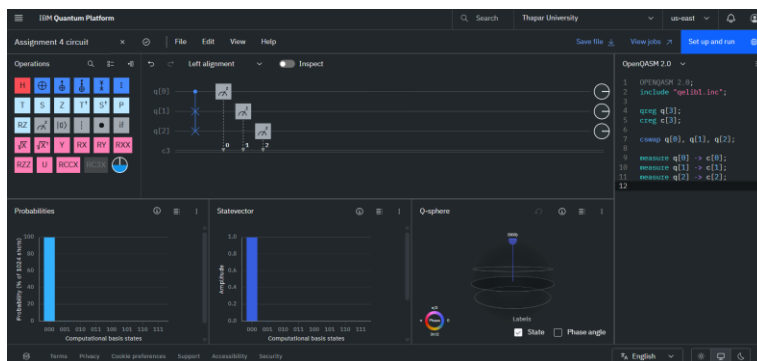


d. State  $|111\rangle$

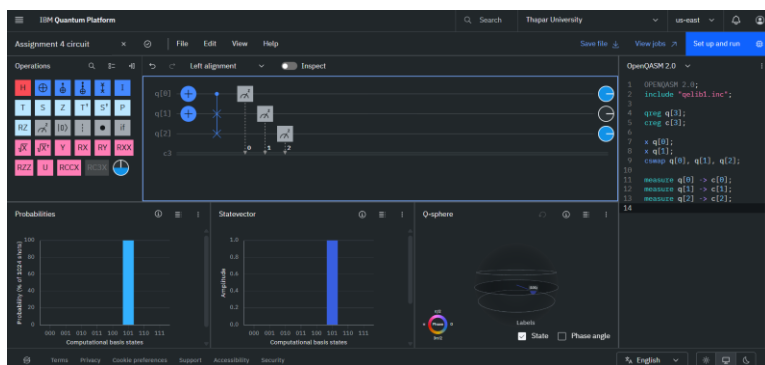


### 3. Fredkin gate (CSWAP) Gate

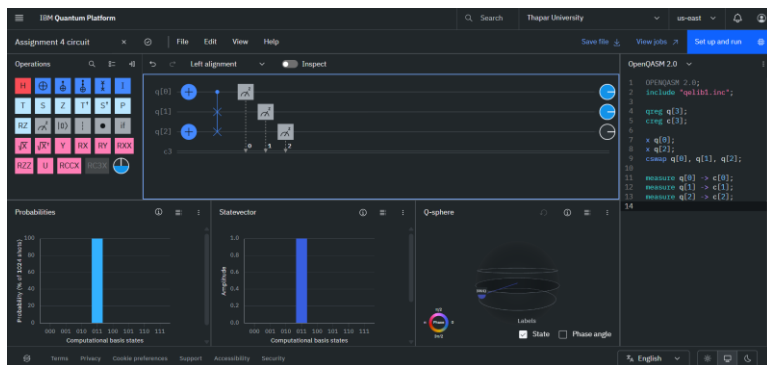
a. State  $|000\rangle$



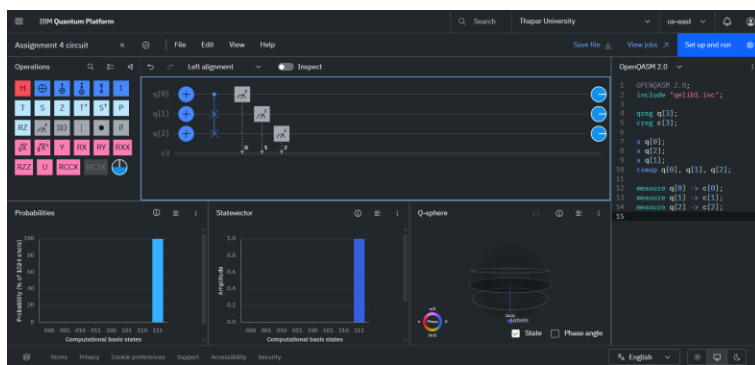
b.  $|110\rangle$



c.  $|101\rangle$

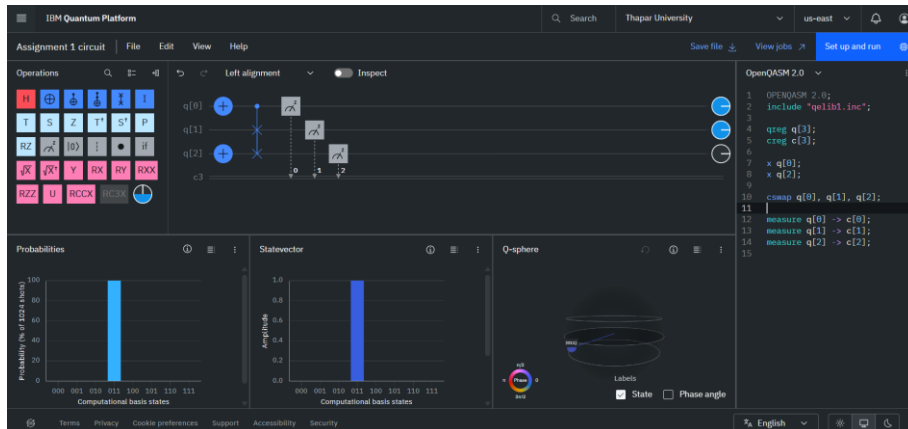


d.  $|111\rangle$



## Experiment 3-:

To show that the CSWAP gate is its own inverse.

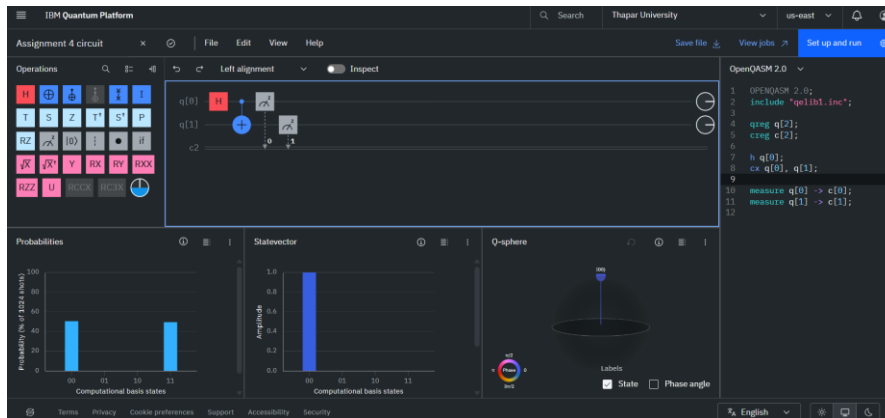




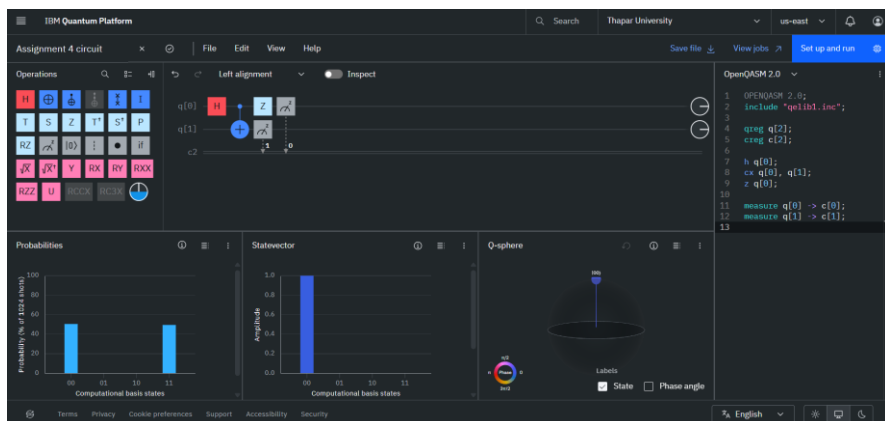
# Experiment 4:-

To generate and analyze all four maximally entangled two-qubit Bell states using appropriate quantum circuits.

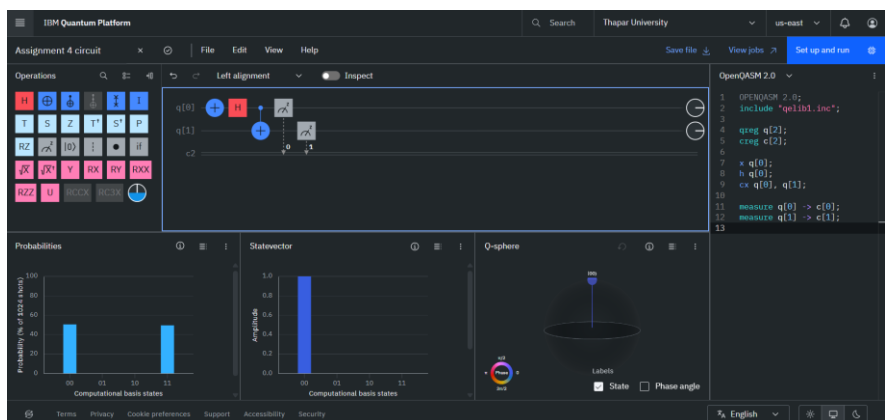
## 1. For $|\Phi^+\rangle$



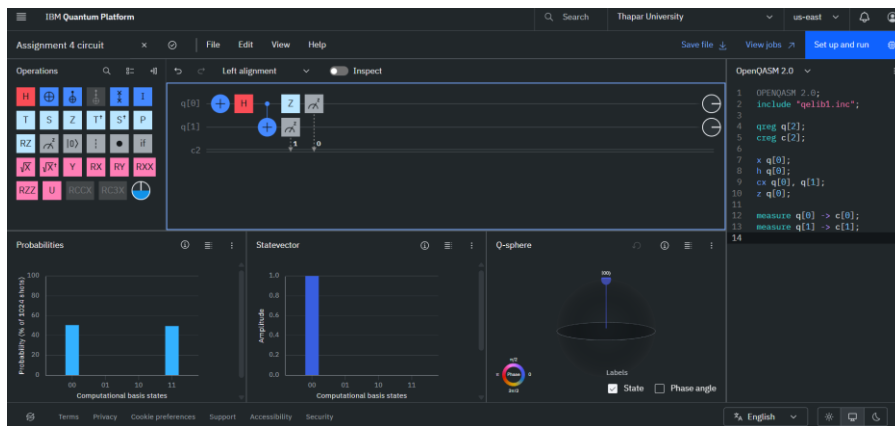
## 2. For $|\Phi^-\rangle$



## 3. For $|\Psi^+\rangle$

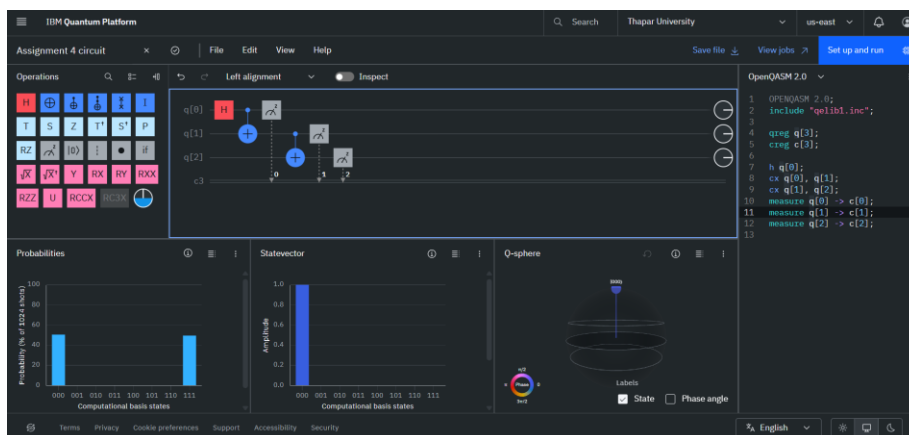


#### 4. For $|\Psi\rangle$



## Experiment 5:-

To generate and analyze a three-qubit maximally entangled Greenberger–Horne–Zeiling (GHZ) state using a quantum circuit.

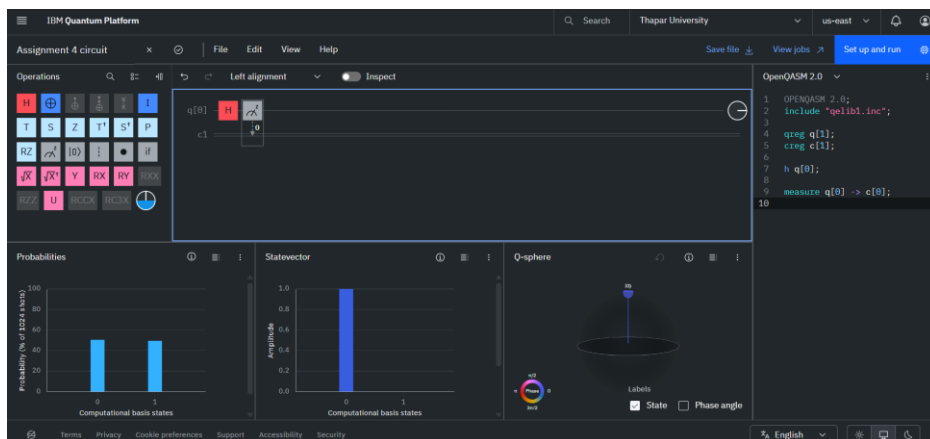
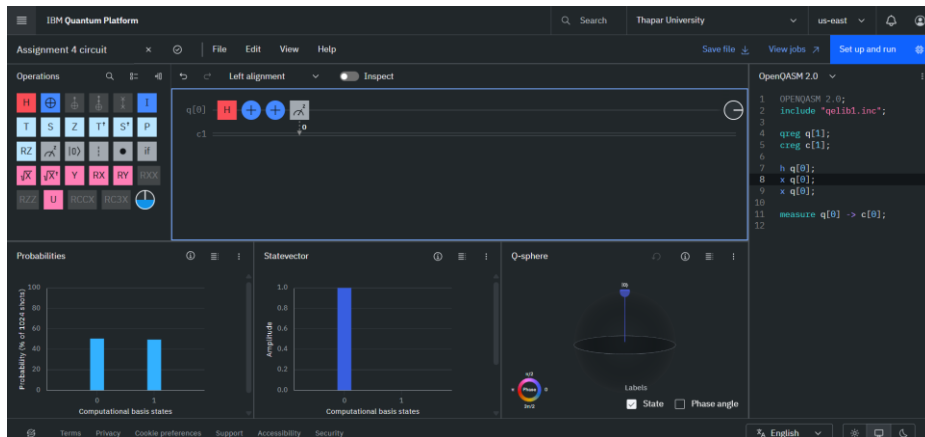


# Experiment 6:-

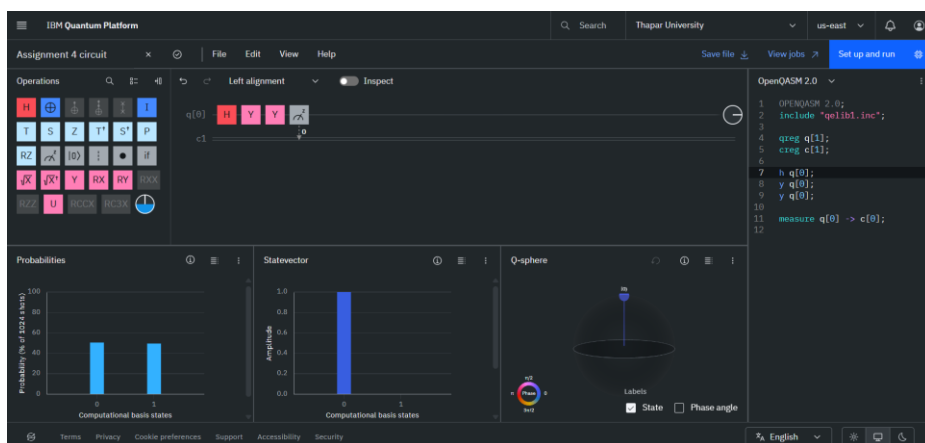
To verify important identities between single-qubit quantum gates by constructing equivalent quantum circuits using OpenQASM and comparing their action on a qubit.

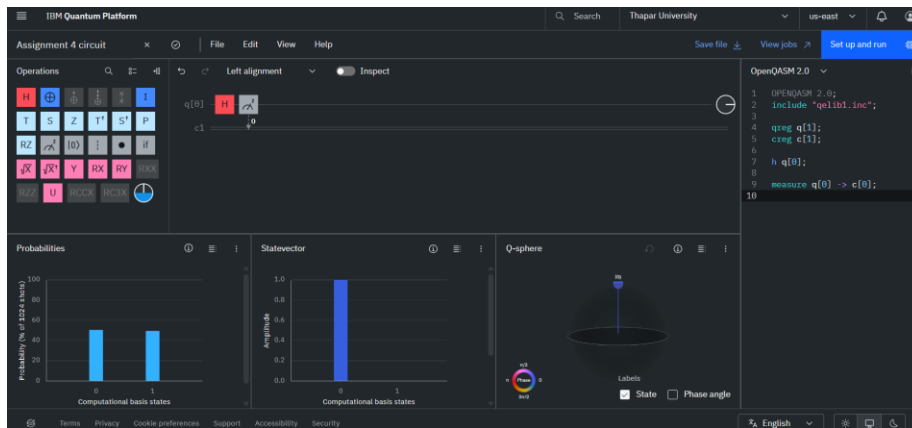
1. For  $X^2 = Y^2 = Z^2 = I$

a.  $X^2 = I$

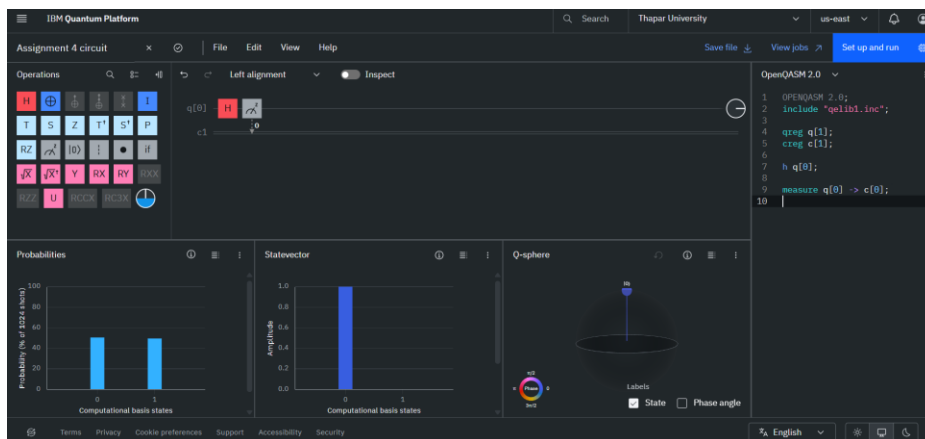
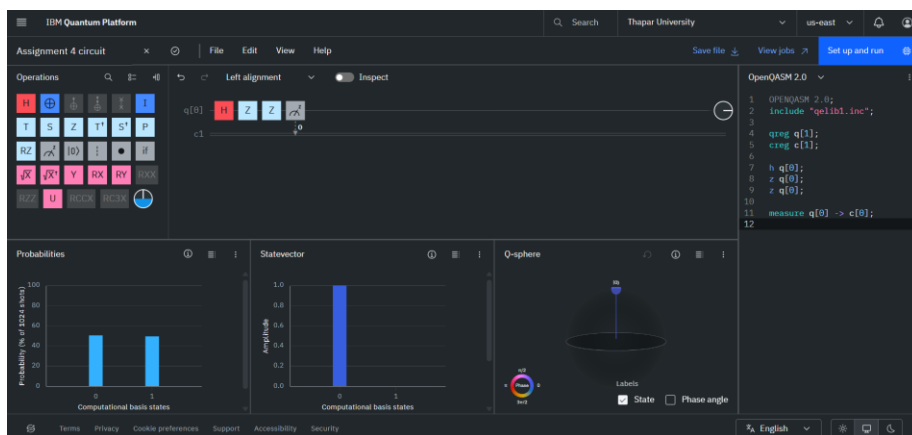


b.  $Y^2 = I$



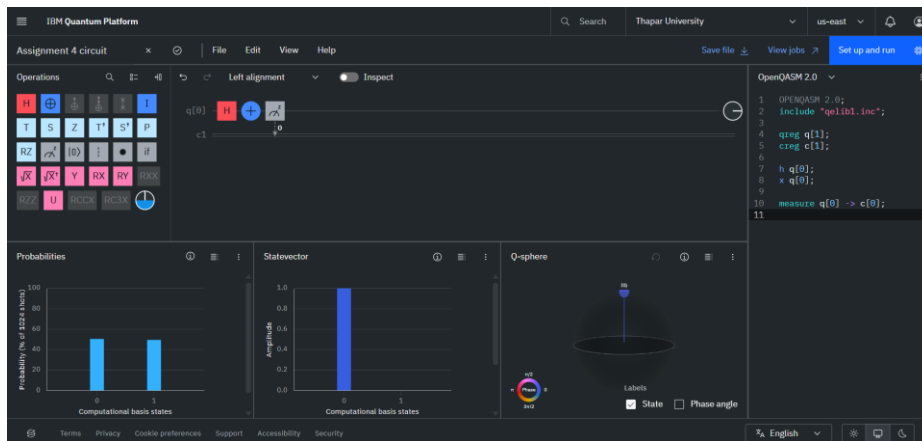


c.  $Z^2=I$

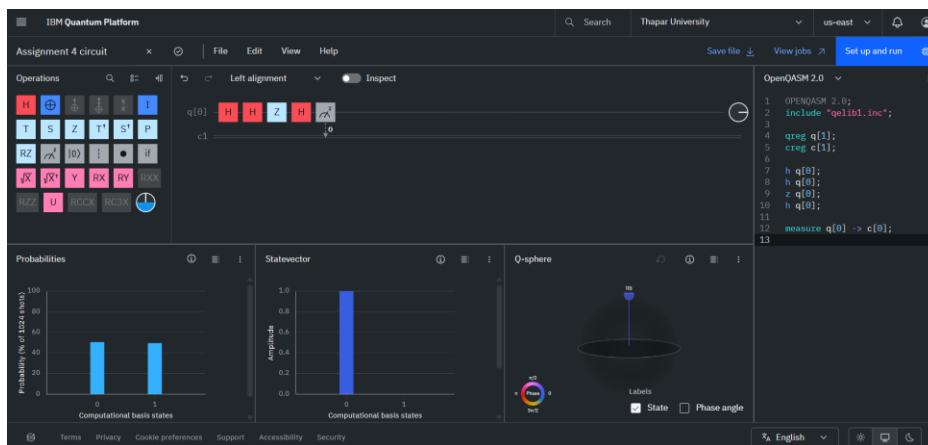


## 2. $X=HZH$

a. For X

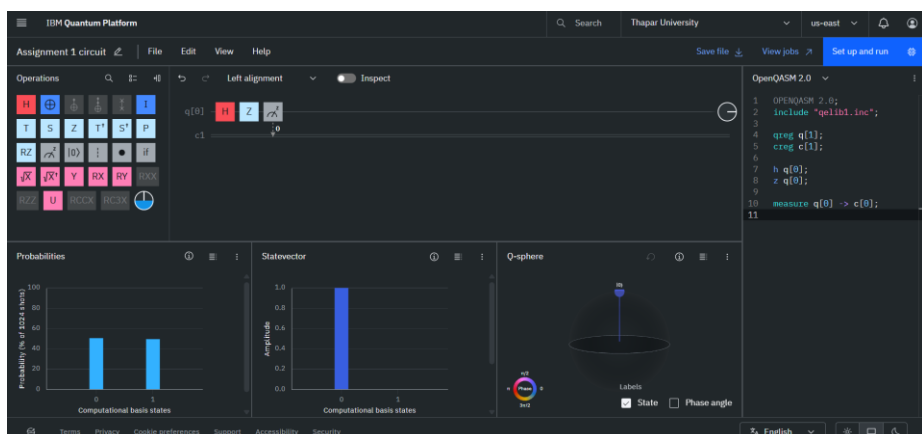


b. For HZH

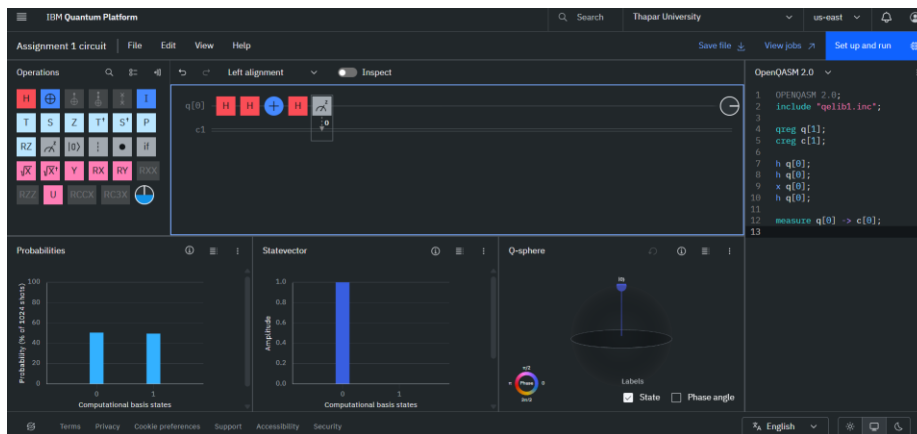


## 3. $Z=HXH$

a. For Z

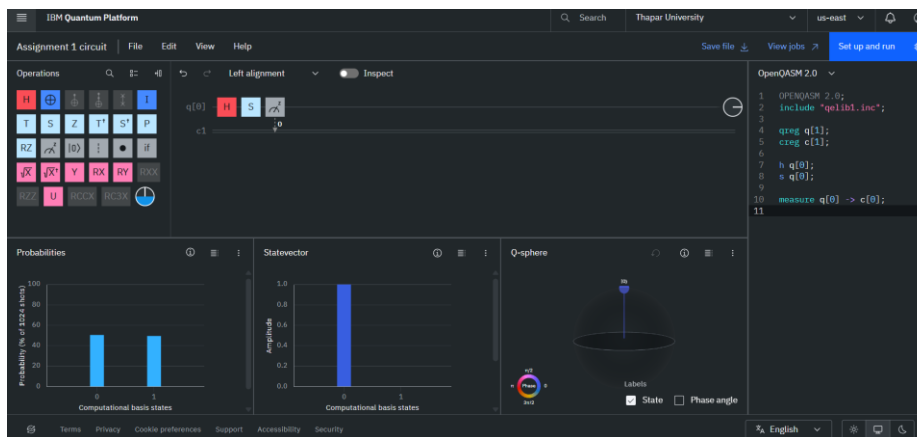


b. For HXH

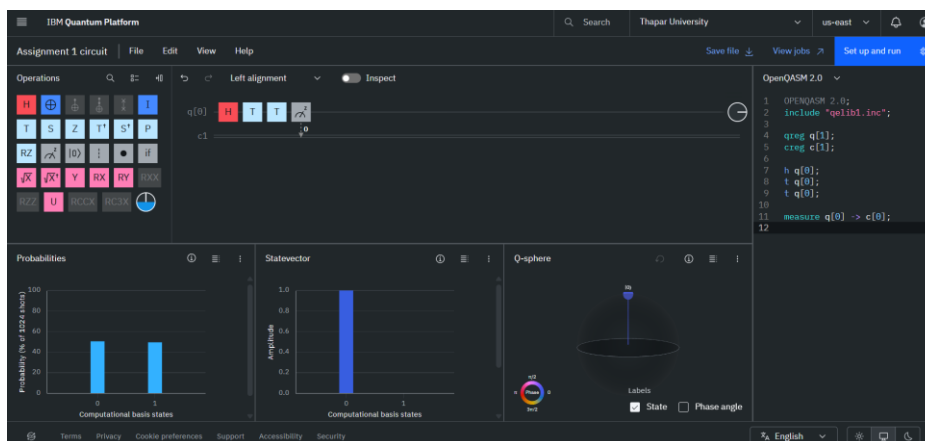


4.  $S=T^2$

a. For S

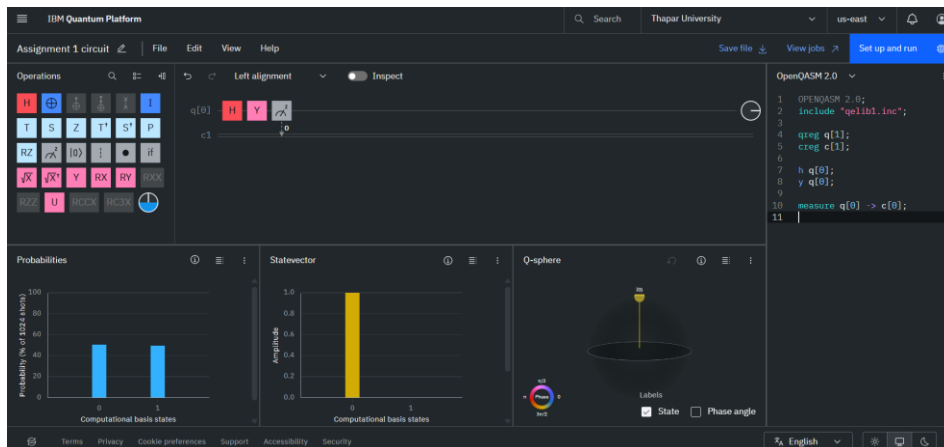


b. For T2

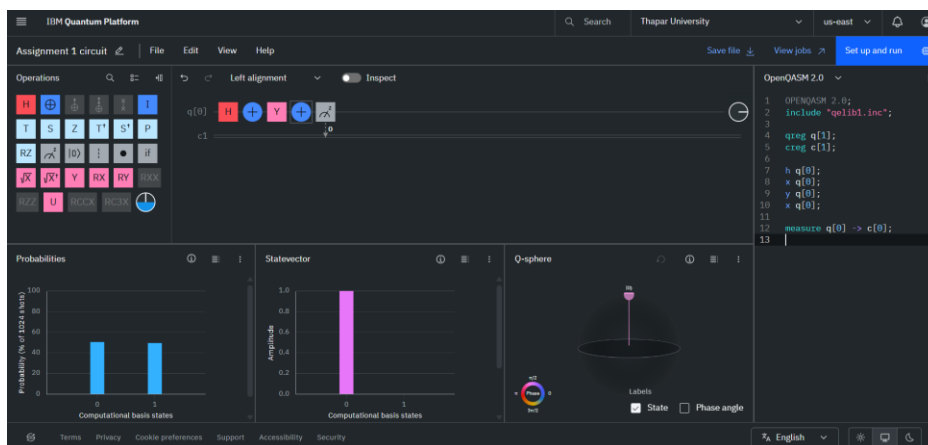


5.  $-iY = XYX$

a. For  $-iY$



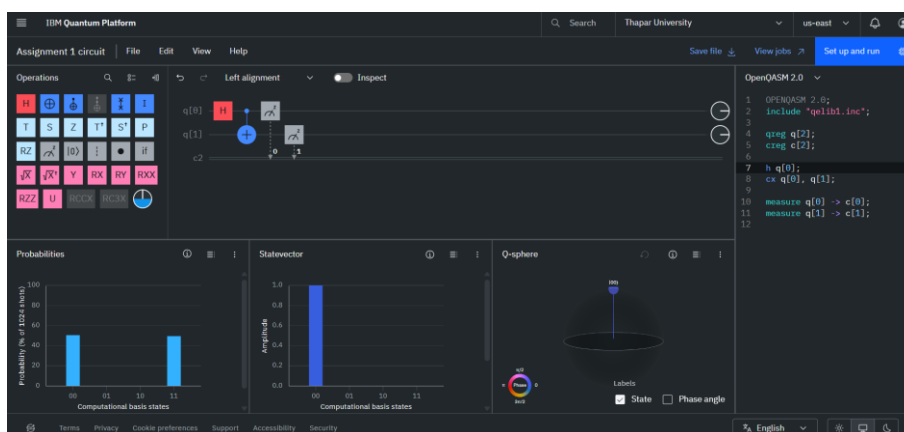
b. For  $XYX$

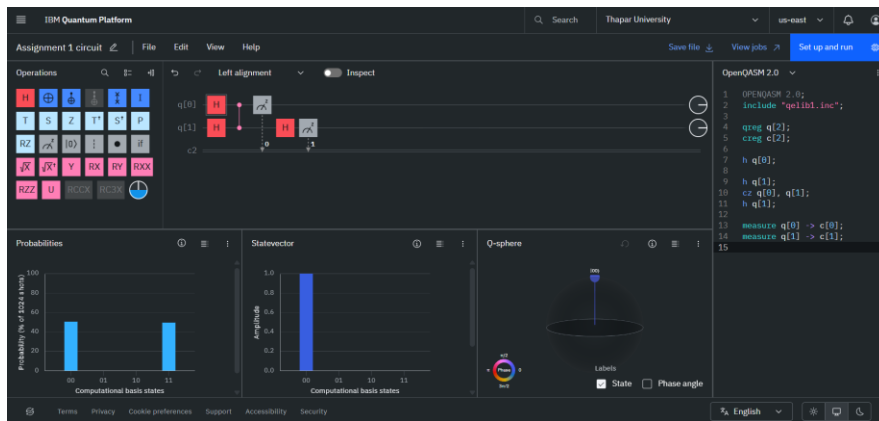


## Experiment 7:-

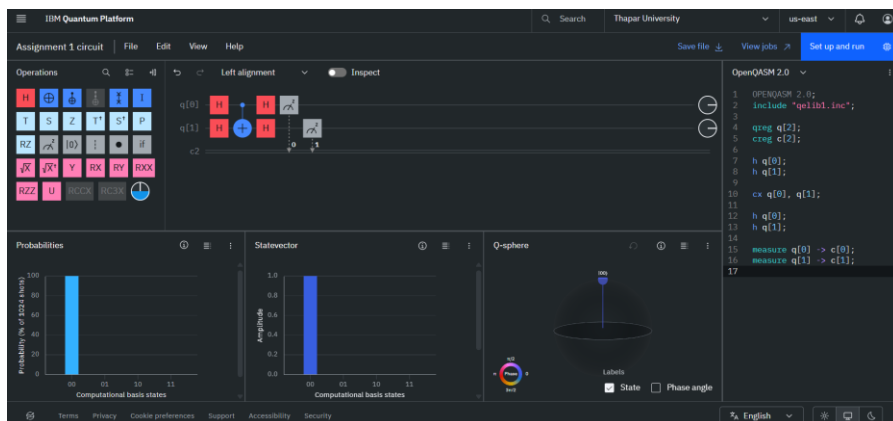
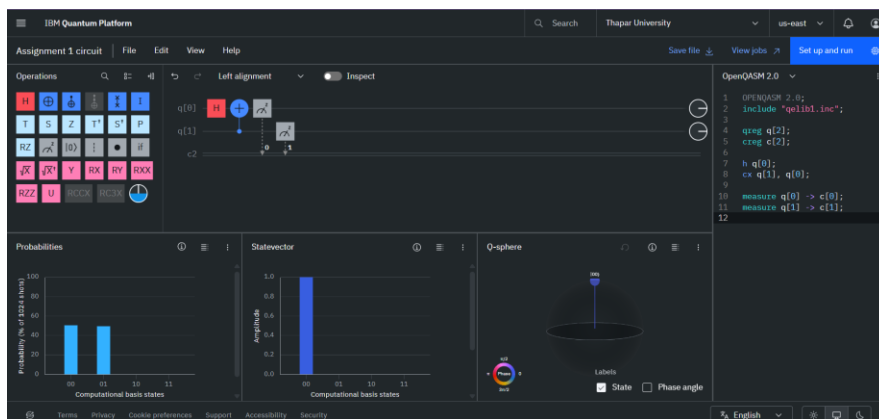
To verify the equivalence of various multi-qubit quantum circuits using OpenQASM and to compare their action on different qubits.

1.

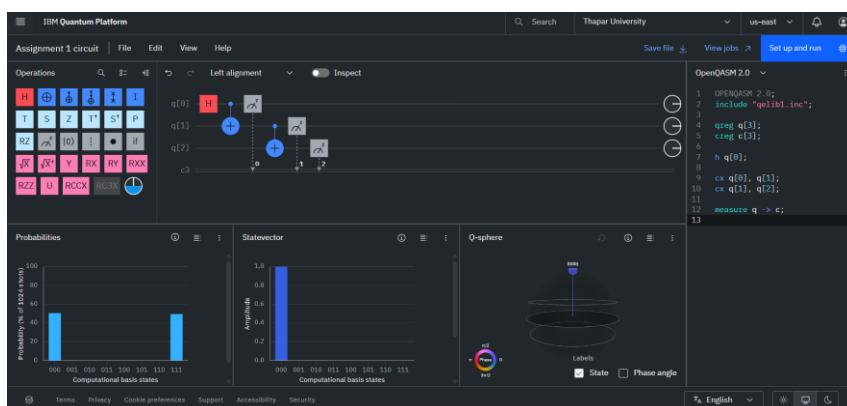




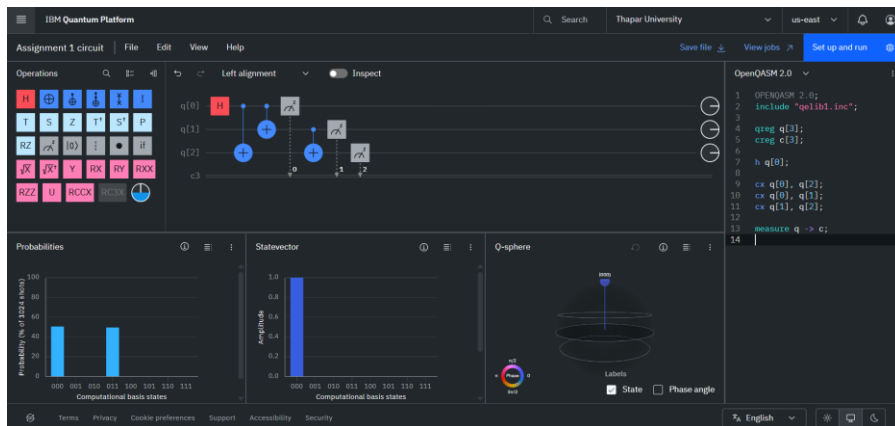
2.



3.







4.

